

DOTE 6635: Artificial Intelligence for Business Research (Spring 2026)

Model-Free Reinforcement Learning

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Core Challenges of RL

- Temporal credit assignment
 - Which earlier actions/decisions deserve "credit" or "blame" for outcomes that happen later?
- Exploration
 - How to efficiently explore to gain information?
- Generalization
 - How policies learned in one environment could generalize?
- RL to solve a large planning problem using a learning approach
 - AlphaGo, AlphaStar, simulated robotics, LLM coding, LLM math, etc.
 - Transition dynamics is known, but with a very very large state space.
 - Run the simulator to collect data.
 - RL proved to be very successful.
- RL to solve a learning problem
 - Adaptive medical treatment, deep research agent, AI scientist, etc.
 - Transition dynamics unknown, reward unknown (and too many states).
 - Interact with the environment to collect data.
 - Great potential for RL, but not realized yet.

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Agenda

- Monte Carlo Method
- Temporal Difference

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Model-Free RL

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-4-model-free-prediction-.pdf>

- So far we assume that we **know all four important sets of information**:
 - Action Space A
 - State Space S
 - Transition Matrix P
 - Reward Function R
- Now, we relax this assumption and assume we **do not know the reward function and transition matrix**.

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Monte-Carlo RL

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-4-model-free-prediction-.pdf>

- Model-free learning:
 - You do not know reward function and transition matrix.
 - However, you can take actions and **experiment with the environment** to observe the outcomes.
- **Goal**: Given a policy, learn the value function from data.
- Monte-Carlo method:
 - MC RL learns directly from **episodes of experience**. All episode must terminate.
 - MC is model-free, without any knowledge of MDP transitions or rewards.
 - MC leverages the simplest possible idea: **value function = mean return**.

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MC Policy Evaluation

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- Objective: Estimate the value function v_π under policy π :

$$S_0, A_0, R_0, \dots, S_T \sim \pi$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T$$

$$v_\pi(s) = \mathbb{E}_\pi [G_t \mid S_t = s]$$

- The idea of Monte Carlo: use empirical mean return to approximate expected return.
- **First-visit MC**: v_π is estimated by the average return following each first visit to a state s in a set of episodes.
- **Every-visit MC**: v_π is estimated by the average return following each visit to a state s in a set of episodes.
- If there is an absorbing state, everything is well-defined; otherwise, we need to specify a finite T , which is larger if γ is larger.

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First-Visit MC Policy Evaluation

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- **Initialization:**
 - N (counter), $N(s) \leftarrow 0$ for all $s \in \mathcal{S}$
 - $\text{Returns}(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$
- **Repeat:**
 - Generate** an episode following policy π
 - For each** distinct s appearing in the episode
 - $G \leftarrow$ return following the first occurrence of s
 - $N(s) \leftarrow N(s) + 1$
 - $\text{Returns}(s) \leftarrow \text{Returns}(s) + G$
- **Output:**
 - For each** distinct s
 - $N^{-1}(s)\text{Returns}(s) \longrightarrow v_{\pi}(s)$ as $N(s) \rightarrow +\infty$

- First-visit evaluation is unbiased, and it converges to the true value function.

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Every-Visit MC Policy Evaluation

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- **Initialization:**
 - $N \leftarrow$ counter, $N(s) \leftarrow 0$ for all $s \in \mathcal{S}$
 - $\text{Returns}(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$
- **Repeat:**
 - Generate** an episode following policy π
 - For each** s appearing in the episode
 - $G \leftarrow$ return following the occurrence of s
 - $N(s) \leftarrow N(s) + 1$
 - $\text{Returns}(s) \leftarrow \text{Returns}(s) + G$
- **Output:**
 - For each** distinct s
 - $N^{-1}(s)\text{Returns}(s) \longrightarrow v_{\pi}(s)$ as $N(s) \rightarrow +\infty$

- Every-visit evaluation is biased, but it still converges to the true value function. And it usually has a smaller MSE because the sample size is much larger.

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Incremental MC Policy Evaluation

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- An incremental way to compute mean value:

$$\begin{aligned}\mu_k &= \frac{1}{k} \sum_{j=1}^k x_j \\ &= \frac{1}{k} \left(x_k + \sum_{j=1}^{k-1} x_j \right) \\ &= \frac{1}{k} (x_k + (k-1)\mu_{k-1}) \\ &= \mu_{k-1} + \frac{1}{k} (x_k - \mu_{k-1})\end{aligned}$$

- Similarly, for MC policy evaluation:

$$\begin{aligned}N(S_t) &\leftarrow N(S_t) + 1 \\ V(S_t) &\leftarrow V(S_t) + \frac{1}{N(S_t)} (G_t - V(S_t))\end{aligned}$$

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Incremental MC Policy Evaluation

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- This incremental update also means less memory usage.
- More importantly, it allows for non-stationary policy (i.e., the transition matrices may change over time):
 - We only consider stationary policy so far, because a stationary MDP with finite states is guaranteed to have a stationary optimal policy.
 - If we go beyond this setting, i.e., policy at time t is different from policy at time $t + 1$.

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

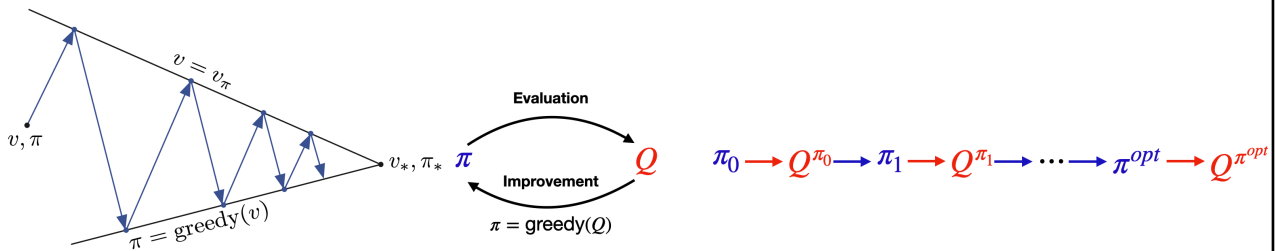
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MC Policy Iteration

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-5-model-free-control-.pdf>

- Objective: use MC estimation to learn the optimal policy.
- MC marries policy iteration:



- Policy Evaluation: Compute v_π via MC policy evaluation.
- Policy Iteration: Improve the current policy w.r.t. the current state-action value function Q_π .
- Challenge: under a policy, many state-action pairs may **never be visited**.
- Solution: **epsilon-greedy**.

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MC Policy Iteration with ϵ -Greedy

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-5-model-free-control-.pdf>

- Initialization:

N (counter), $N(s, a) \leftarrow 0$ for all $s \in \mathcal{S}$, $a \in \mathcal{A}$
Returns(s, a) $\leftarrow$ empty lists, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$
 $\pi \leftarrow$ arbitrary ϵ -greedy policy
 $Q \leftarrow$ arbitrary

- Repeat:

Generate an episode using exploring starts and policy π
For each distinct (s, a) appearing in the episode
 $G \leftarrow$ return following the first occurrence of (s, a)
 $N(s, a) \leftarrow N(s, a) + 1$
Returns(s, a) $\leftarrow$ **Returns**(s, a) + G
 $Q(s, a) \leftarrow$ **Returns**(s, a) / $N(s, a)$
For each distinct s :

$$\pi(a|s) \leftarrow \begin{cases} \epsilon/m + 1 - \epsilon, & \text{if } a = \arg \max Q(s, a) \\ \epsilon/m, & \text{otherwise} \end{cases}$$

Thm: The ϵ -greedy policy iteration π' with respect to q_π is an improvement over any ϵ -greedy policy π , i.e.,

$$v_{\pi'}(s) \geq v_\pi(s)$$

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Temporal-Difference Learning

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- "If one had to identify one idea as central and novel to RL, it would undoubtedly be temporal-difference (TD) learning." - Sutton and Barto (2017)
- TD is a combination of MC and Bellman operator.
- TD is a model-free method.
- TD can be used in episodic or non-episodic settings.
- TD updates a guess towards a guess.

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Temporal-Difference Learning

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- Recall that, given a policy π , the value function can be obtained as:

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 r_{t+3} + \dots \text{ in MDP } M \text{ under policy } \pi$$

$$V^\pi(s) = \mathbb{E}_\pi[G_t | s_t = s]$$

- Bellman operator and value iteration:

$$B^\pi V(s) = r(s, \pi(s)) + \gamma \sum_{s' \in S} p(s' | s, \pi(s)) V(s')$$

- The incremental MC: $V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$

■ Simplest temporal-difference learning algorithm: TD(0)

- Update value $V(S_t)$ toward *estimated* return $R_{t+1} + \gamma V(S_{t+1})$

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

TD can be thought of as a **bootstrap method** to update based on an existing estimate.

- $R_{t+1} + \gamma V(S_{t+1})$ is called the *TD target*
- $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$ is called the *TD error*

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TD(0) Algorithm

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- Input:** π policy to be evaluated, step size α

- Initialization:** V arbitrary

- Repeat** for each episode:

Initialize state s

Repeat for each step of the episode:

$a \leftarrow$ action given by π for s

Take action a , observe reward r and next state s'

$V(s) \leftarrow V(s) + \alpha [r + \gamma V(s') - V(s)]$

$s \leftarrow s'$

until s is a terminal state

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MC vs. TD

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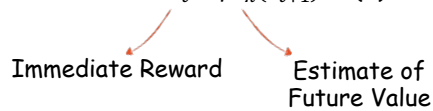
- Recall under the MDP model:

$$V^\pi(s) = \mathbb{E}^\pi(G_t | S_t = s) \quad (1)$$

$$= \mathbb{E}^\pi\left(\sum_{k=0}^{\infty} \gamma^k R_{t+k} | S_t = s\right)$$

$$= \mathbb{E}^\pi[R_t + \gamma V^\pi(S_{t+1}) | S_t = s] \quad (2)$$

- MC methods use G_t in (1) as the target.
- TD methods use $R_t + \gamma v_\pi(S_{t+1})$ in (2) as the target.



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MC vs. TD: Pros and Cons

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- | | |
|---|---|
| • MC must wait until the end of an episode. | • TD can learn online after each step. |
| • MC learns from complete sequences. | • TD can learn from incomplete sequences. |
| • MC only works for episodic environments. | • TD works in continuing environments. |

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MC vs. TD: Bias-Variance Trade-off

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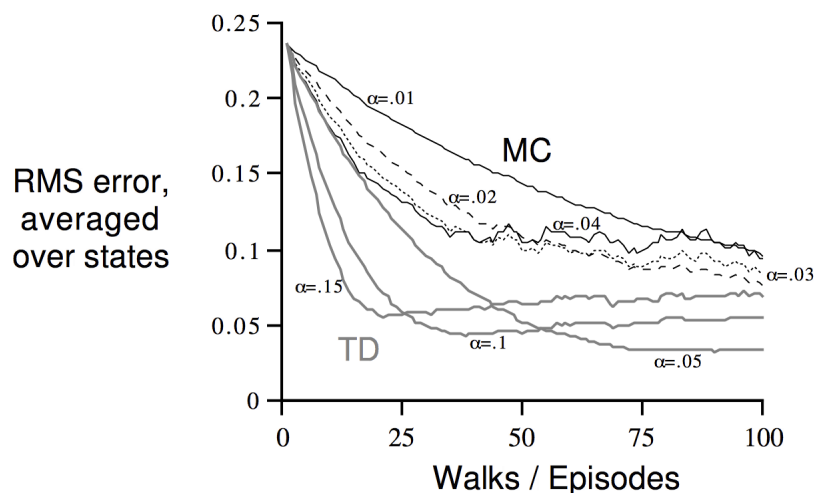
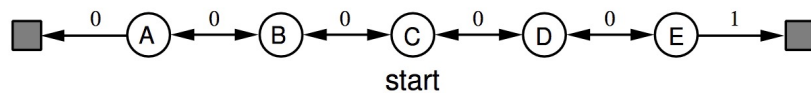
- MC updates on G_t after each episode, which is an **unbiased estimate** of $v_\pi(S_t)$.
- TD updates on $R_t + \gamma v(S_{t+1})$, not $R_t + \gamma v_\pi(S_{t+1})$, which is a **biased estimate** of $v_\pi(S_t)$.
- MC has high variance, zero bias.
 - Good convergence even with **function approximation** (using, e.g., DNN).
 - **Not sensitive** to initial values.
 - Simple idea **without even knowing Bellman equation**.
 - More efficient in **non-Markov environments**.
- TD has much lower variance, some bias.
 - Return depends on **many** random actions/transitions/rewards.
 - TD target only depends on **one** random action/transition/reward.
 - **More efficient** than MC, because it exploits Markov property and the Bellman equation.
 - TD(0) converges (but not always with function approximations).
 - **More sensitive to initial values**.

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MC vs. TD: An Example

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